# Results

**Prepared by**: Armeen Shasti-Nazem (University of Washington, aashasti@uw.edu) — independent reimplementation of Newton et al. 2026 (*ApJ*, 1006:184, "Intermediate-mass Black Hole Formation from Hierarchical Mergers in Galactic Nuclei," hereafter N26).

This is the single consolidated results file — validation, both of N26's Section 5.4 extensions, and the headline numbers behind each, in one place. It replaces browsing `phase3_validation_2026-07-26.md`, `phase4_mass_threshold_scan_2026-07-27.md`, and `phase5_smbh_mass_scan_2026-07-28.md` individually; those files (plus the raw per-seed data in `phase*_raw/`) still exist underneath this one for anyone who wants the full diagnostic trail. Every number below is reported under **both** readings of the Eq. 22 $\langle M_{\rm avg} \rangle$ ambiguity (see `paper/limitations_summary.md` §2.1) — it is the single choice that changes almost everything that follows.

## Headline

- **Validation**: the bulk IMBH-formation statistic (% of population > 100 $M_\odot$) reproduces N26's Table 1 to within a few points across all four initial conditions. The extreme upper-mass tail does not — our heaviest initial condition (H18) produces BHs up to **58,000 $M_\odot$** on average vs. Table 1's **407 $M_\odot$** ceiling.
- **Is the critical-mass threshold sharp or gradual?** Gradual. Under the star-only Eq. 22 reading, the probability that *any* BH crosses 100 $M_\odot$ rises smoothly from 0% to 75% across $m_{\rm max} \approx$ 20–32 $M_\odot$, not saturating even at the top of that band. Under the BH-inclusive reading, no threshold appears anywhere in the tested range.
- **Does it generalize to other SMBH masses?** Yes, conditionally. Under star-only, **56 of 56** runs across 3 decades of SMBH mass produced at least one IMBH. Under BH-inclusive, **0 of 56** exceeded 0.5% of the population. A sharp side finding: a runaway-growth failure mode, absent at the Milky Way's own mass, appears in **50–88%** of runs 3–10× below it.

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## 1. Validation against Table 1

$N = 1000$ BHs, $M_\bullet = 4 \times 10^6\, M_\odot$, 10 Gyr, 3 seeds/IC, all four of N26's initial conditions.

| IC | Mergers (ours / Table 1) | Max mass, $M_\odot$ (ours / Table 1) | % > 100 $M_\odot$ (ours / Table 1) | EMRI % (ours) |
|---|---:|---:|---:|---:|
| K20 | 24.0 / 34 | 39.1 / 28.4 | 0.0% / 0% | 29.6% |
| K20+M | 28.7 / 30 | 46.2 / 57.8 | 0.0% / 0% | 31.9% |
| H18 | 1022.0 / 371 | 58,034 / 407.3 | 12.5% / 7.8% | 34.7% |
| H18+M | 1046.7 / 535 | 11,881 / 526.0 | 12.8% / 14.3% | 35.0% |

**K20 and K20+M reproduce well** — mergers and max mass both within ~1.3–1.5× of Table 1, consistent with ordinary seed variance and the K20 mass-function reconstruction uncertainty (`paper/limitations_summary.md` §2.4).

**H18 and H18+M: the bulk statistic matches, the extreme tail doesn't.** The population-level metric that matters most for the paper's headline claim — % of BHs exceeding 100 $M_\odot$ — is close (12.5% vs. 7.8%; 12.8% vs. 14.3%, the latter within seed noise). But max mass is off by 20–220×, driven by 1–3 BHs per 1000-BH run undergoing 60–230 merger generations (Table 1's own max is 12–16G) via a genuine runaway positive-feedback loop in both growth channels — traced mechanistically, not a coding defect (full trace: `paper/limitations.md#phase2-emri-rate-high`). The 99th- percentile BH mass (677–851 $M_\odot$ for H18) is much closer to Table 1's scale than the single most extreme outlier is.

**EMRI rate runs hot across all four ICs**: 30–37% over 10 Gyr here vs. N26's implied ~4–5% (from their

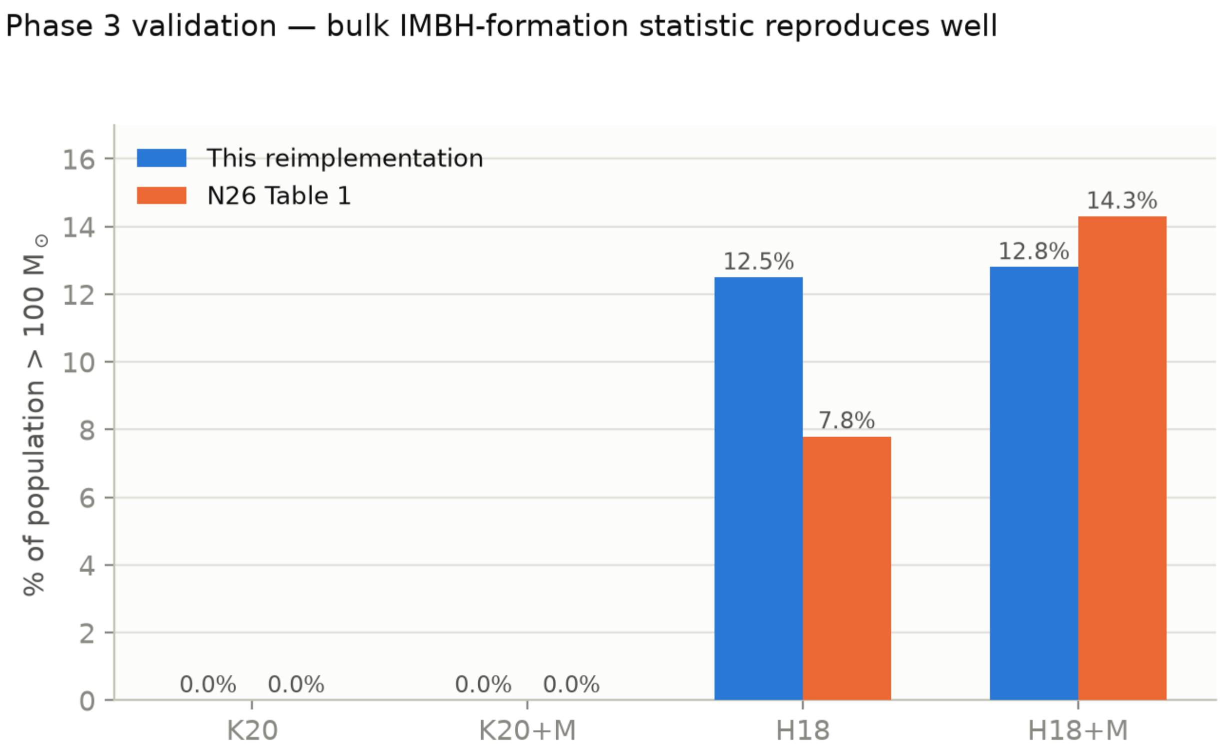


Figure 1: Phase 3 validation

~4–4.8 Gyr$^{-1}$ merger-rate figures, our conversion). Still an open item — see `paper/limitations_summary.md` §2.1.

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## 2. Is the critical initial-mass threshold sharp or gradual?

Log-uniform mass family on $[6, m_{\max}]\,M_\odot$, $m_{\max}$ scanned 16–100 M$_\odot$, $N = 1000$, 10 Gyr, both Eq. 22 readings. Filled markers below are 8-seed points (refined pass); open markers are the original 3-seed pass.

| $m_{\max}$ [M$_\odot$] | Any IMBH? (star-only) | 95% CI (Wilson) |
|---|---|---|
| 20.1 | 0/8 (0%) | 0–32% |
| 22.6 | 3/8 (38%) | 14–69% |
| 25.3 | 4/8 (50%) | 22–79% |
| 28.4 | 6/8 (75%) | 41–93% |
| 31.8 | 6/8 (75%) | 41–93% |

**Under star-only**, this is a genuine crossover, not a step function: even at $m_{\max} = 31.8$ M$_\odot$ — which a thinner 3-seed pass made look fully saturated — 2 of 8 seeds still produce zero IMBHs. Wilson CIs overlap substantially between adjacent grid points, so the data supports "onset somewhere in ~20–32 M$_\odot$, 50% point closer to 23–26 M$_\odot$" but not a tighter claim. Beyond 40 M$_\odot$, all tested points (3 seeds each) saturate at 3/3 — consistent with, but not independently re-verified at, 8 seeds.

**Under BH-inclusive**, no threshold appears anywhere in the range: flat 0.0% from 16–79.5 M$_\odot$, only 0.1–0.3% even at $m_{\max} = 100$ (H18 exactly).

**The primordial-binary-merger axis** (N26's own 0%/15% "+M" prescription, tested at the same 5 grid points, 8 seeds each) has **no detectable effect** on the crossover: pooled 19/40 (0%) vs. 20/40 (15%) any-IMBH runs, Fisher's exact $p = 1.0$. The one clean effect is a **+2.3%** mean-mass bump at every grid point — matching Phase 3's real K20/K20+M (+2.2%) and H18/H18+M (+2.4%) shifts almost exactly, a good cross-check that the extension behaves correctly. Mechanism: primordial mergers touch only ~2.5% of the population directly, a far smaller lever than one step of the $m_{\max}$ grid (~12% multiplicative).

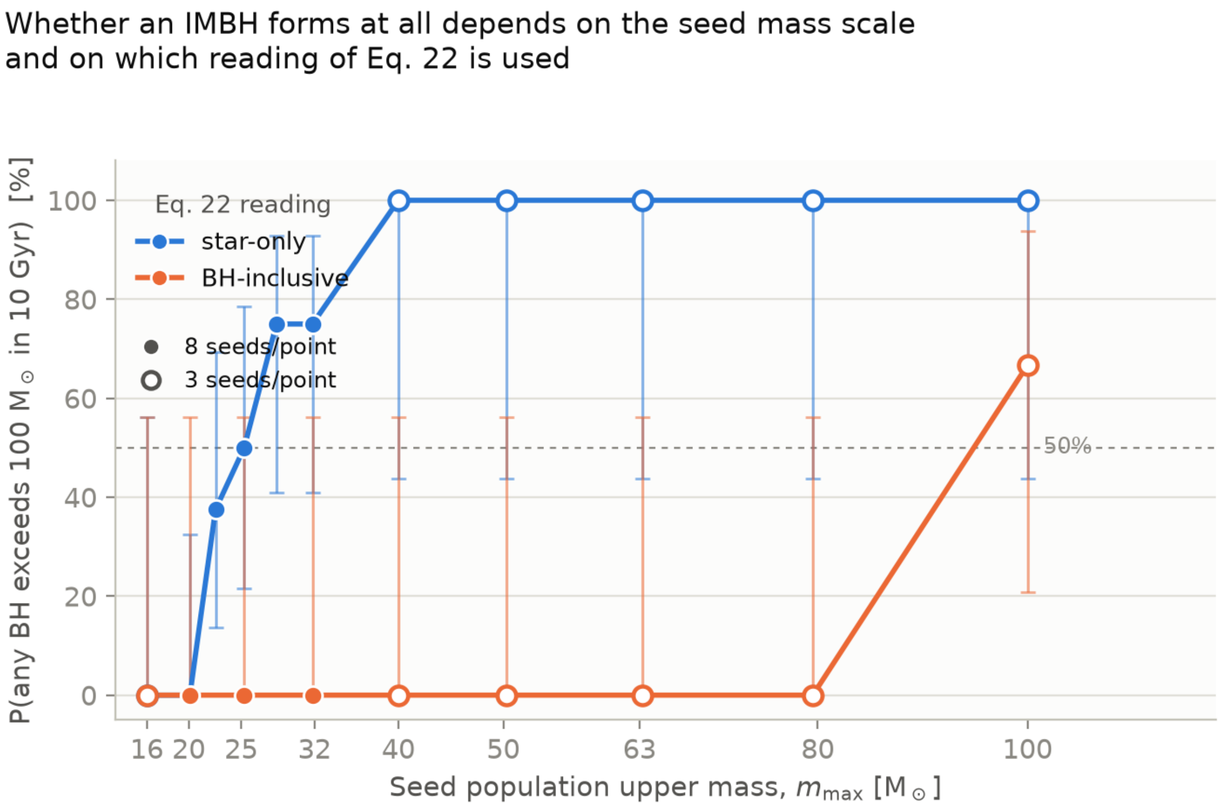


Figure 2: Phase 4 crossover

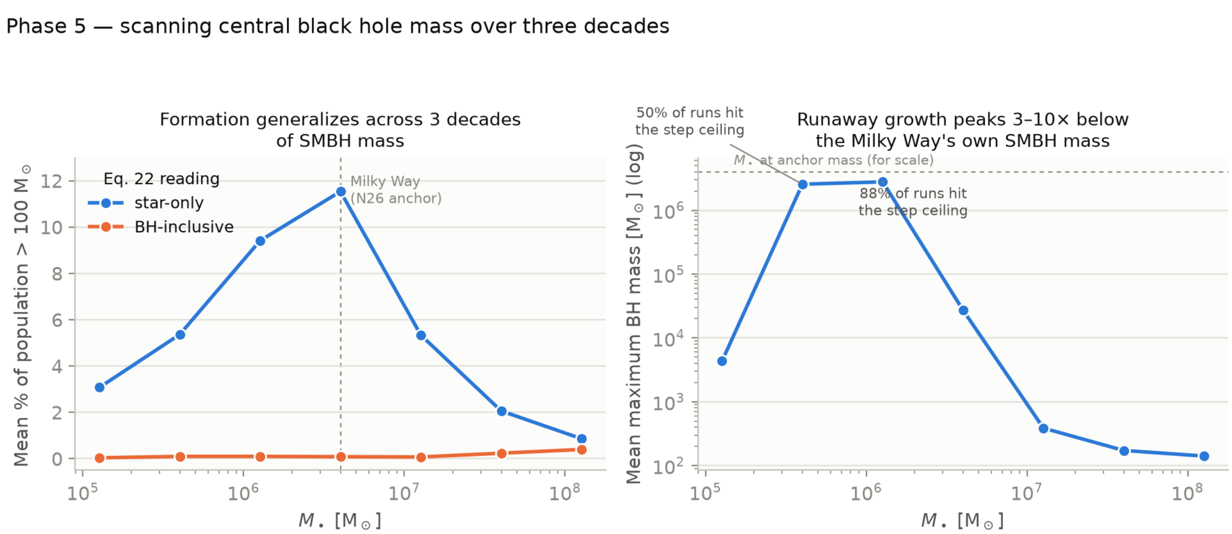


Figure 3: Phase 5 SMBH-mass scan

**Finding 2 — a severe, non-monotonic runaway-growth "sweet spot" 3–10× below the anchor mass.**

| $m_{\mathrm{smbh}}$ [$M_\odot$] | Ratio to anchor | Step-ceiling hit rate | Mean max BH mass [$M_\odot$] |
|---:|---:|:---:|---:|
| $1.26 \times 10^5$ | 0.03× | 0/8 (0%) | 4,326 |
| $4 \times 10^5$ | 0.1× | 4/8 (50%) | 2,555,496 |
| $1.26 \times 10^6$ | 0.32× | 7/8 (88%) | 2,782,188 |
| $4 \times 10^6$ (anchor) | 1× | 0/8 (0%) | 27,132 |
| $1.26 \times 10^7$–$1.26 \times 10^8$ | 3–32× | 0/8 (0%) | 141–386 |

In this band, individual BHs reach into the **millions of solar masses** — in several runs, literally exceeding the central SMBH's own mass. Mechanism: relaxation-driven EMRI removal weakens as $m_{\mathrm{smbh}}$ drops ($t_{\mathrm{relax}} \propto M_\bullet^{1.5}$) while both growth channels' efficiency simultaneously rises ($\propto \sigma^{-2}$-type scalings) — the two effects cross in this specific band rather than one dominating everywhere, which is why the lowest mass tested ($1.26 \times 10^5$, where EMRI removal wins outright at 97.7%) is *not* where growth is worst.

**Explicitly scoped**: this scan holds the cluster's structural density profile fixed at the Milky Way's own values while varying only $m_{\mathrm{smbh}}$ — our own extension, not from N26, made to isolate the pure dynamical effect of SMBH mass. A real lower-mass galactic nucleus plausibly has correspondingly lower density too, which this scan doesn't model — so Finding 2 is a property of this specific scan design, not a claim about real low-mass nuclei.

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## Reproducing these numbers

| Phase | Script | Raw data |
|---|---|---|
| 3 — validation | `scripts/phase3_validation.py` | `results/phase3_validation_2026-07-26_raw.c` |
| 4 — mass threshold | `scripts/phase4_mass_threshold_scan.py`, `phase4b_threshold_refinement.py`, `phase4c_primordial_binary_check.py` | `results/phase4_raw/`, `phase4b_raw/`, `phase4c_raw/` |
| 5 — SMBH-mass scan | `scripts/phase5_smbh_mass_scan.py` | `results/phase5_raw/` |

Full per-decision reasoning (why each grid, seed count, and Eq. 22 reading was chosen) is in `paper/limitations_summary.md`; the complete session-by-session log is in `paper/limitations.md`.